Explainability in Machine Learning (AI5120)

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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

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So far in

- Introduced Explainability
  - What, Why and When
- Notions surrounding it
  - Taxonomy, Scope, Evaluation, etc.
- Inherently interpretable models
  - Linear models, Rule based, Prototype based, etc.

Now

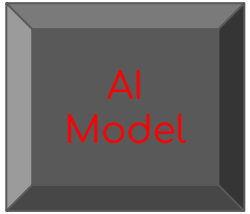
- Post-hoc explanations
  - Local Model-agnostic explanations
    - LIME (Ribeiro et al. SIGKDD 2016) [~24K citations]

Explanation?

- (Human) Interpretable description of the model behavior

Explanation

Understandable



Explanation

Faithful

User

Credits: Prof. Hima Lakkaraju

Explanation

Explanation

AI Model

Model parameters?

Example predictions?

Summarize with a program/rule/tree?

Most important features?

Describe how to flip the prediction?

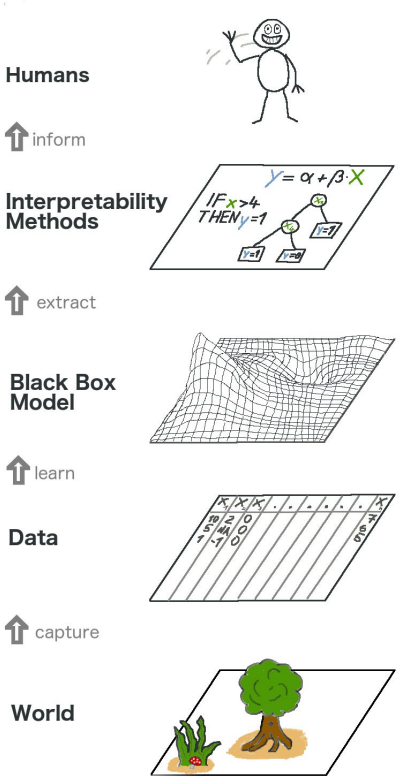
User

Credits: Prof. Hima Lakkaraju

Model-Agnostic Methods

- Separate explanations from the learning model
- Advantage: Flexibility
  - Developers are free to use any model given the interpretation methods are applicable on any
  - Generally, organizations develop multiple models
  - UI becomes independent of the underlying model

Model-Agnostic Methods



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Model-Agnostic Methods

1. Local
   a. Interpret individual predictions
   b. Unearth biases in the local neighborhood of a given instance
2. Global
   a. Describe how features affect the prediction on average
   b. Unearth big picture issues

Local Explanations

Local Explanations

- Feature Importances
- Saliency maps
- Prototypes based
- Counterfactuals
- etc.

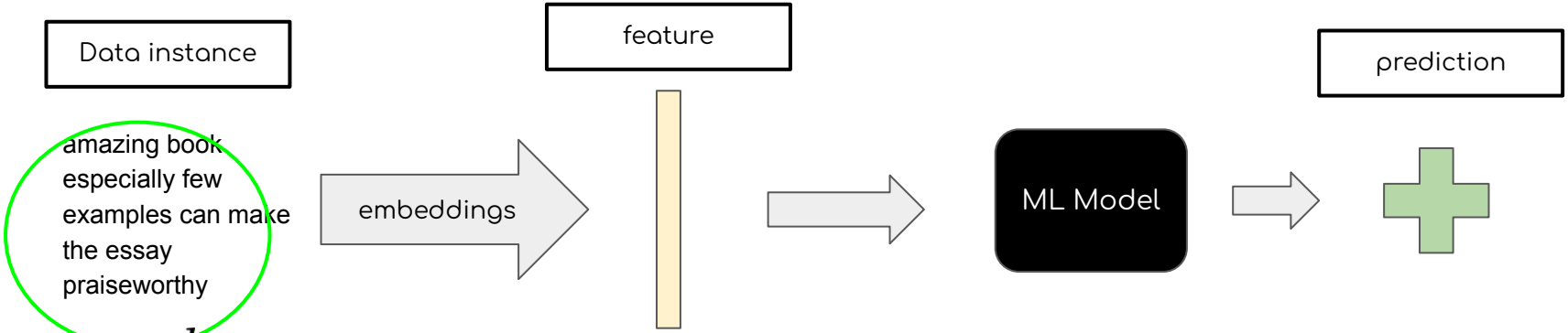
Local Interpretable Model-Agnostic Explanations (LIME)

- Goal is to identify
  - an interpretable model
  - over the interpretable representation
  - That is locally faithful

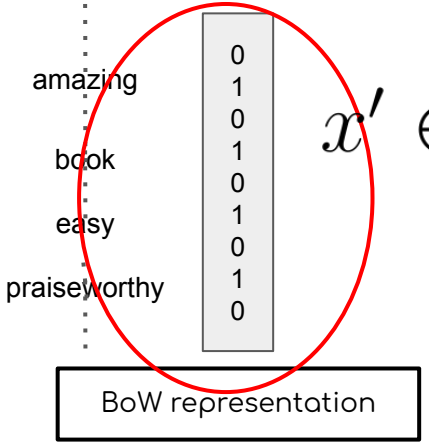
Interpretable data representations

- Interpretable explanations need understandable representations
  - Regardless of the input components or features used by the model

Interpretable data representations



$x \in \mathcal{R}^d$



$x' \in \{0,1\}^{d'}$ Text classification

LIME

Model to be explained $f : \mathcal{R}^d \rightarrow \mathcal{R}$

Explanation is a model $g \in G$ Class of interpretable models

Proximity measure b/w an instance z and x $\pi_x(z)$

Faithfulness of g in the locality $\mathcal{L}(f, g, \pi_x)$ Locality awareness loss

LIME

Measure of complexity

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \quad \mathcal{L}(f, g, \pi_x) + \boxed{\Omega(g)}$$

To ensure both interpretability and local fidelity, minimize L while having Ω to be low

Sampling for local exploration

- LIME is model agnostic
- In order to 'locally' approximate any f, samples need to be drawn
  - Weighted by their proximity
- Sample around x' (interpretable representation)
  - Nonzero elements are drawn uniformly at random
  - Number of the draws are also uniformly sampled

Sampling for local exploration

Sampling gives perturbed sample $z' \in \{0,1\}^{d'}$

Recover the sample in original representation $z \in \mathcal{R}^d$

$f(z)$

label

z'

$\mathcal{Z}$

Dataset of perturbed samples

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Optimize for the local surrogate

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \quad \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

Sparse Linear Explanations

G is the family of linear models

$$g(z') = w_g \cdot z'$$

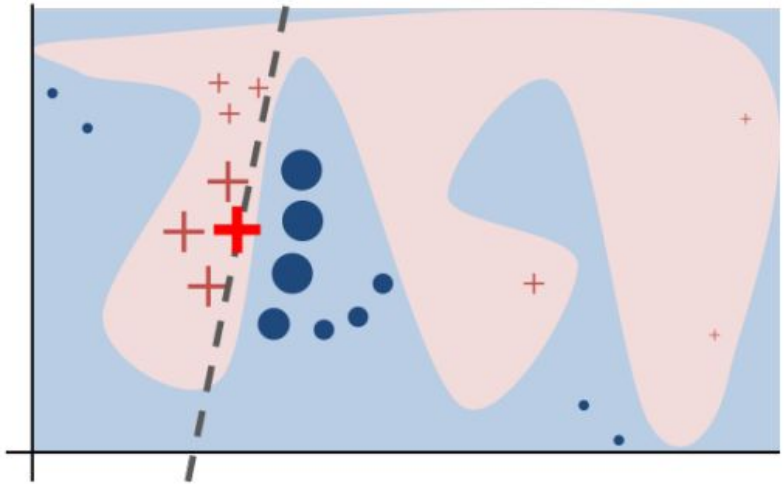
Exponential kernel for proximity

$$\pi_x(z) = e^{-D(x,z)^2/\sigma^2}$$

Weighted square loss

$$\mathcal{L}(f, g, \pi_x) = \sum_{z,z' \in \mathcal{Z}} \pi_x(z) \left(f(z) - g(z')\right)^2$$

Toy example visualization



Sparse Linear Explanations

- Interpretable models need to perform variable selection

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \quad \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

$$\Omega(g) = \infty \mathbb{1}[\|w_g\|_0 > K]$$

Intractable to optimize → Select K features with stepwise regression then learn the weights (K-Lasso)

Sparse Linear Explanations

Algorithm 1 Sparse Linear Explanations using LIME

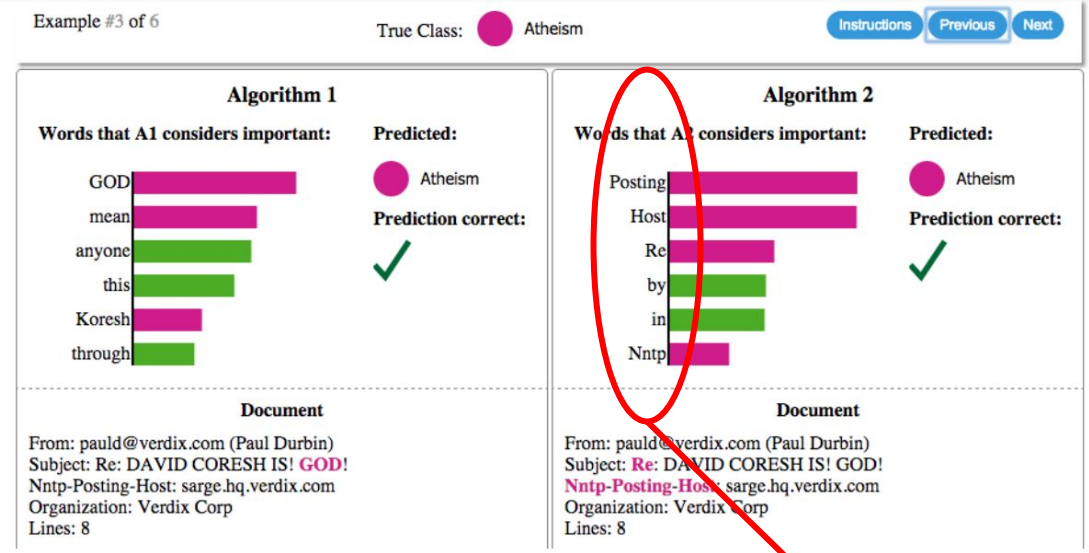
Require: Classifier f, Number of samples N
Require: Instance x, and its interpretable version x'
Require: Similarity kernel π_x, Length of explanation K

$\mathcal{Z} \leftarrow \{\}$
for $i \in \{1, 2, 3, ..., N\}$ **do**
  $z'_i \leftarrow sample_around(x')$
  $\mathcal{Z} \leftarrow \mathcal{Z} \cup \langle z'_i, f(z_i), \pi_x(z_i) \rangle$
end for
$w \leftarrow$ K-Lasso$(\mathcal{Z}, K)$ ▷ with z'_i as features, $f(z)$ as target
return w

Example 1: Text classification

- SVM with RBF kernel
- Christianity vs. Atheism
- Subset of 20 newsgroup dataset

Example 1: Text classification

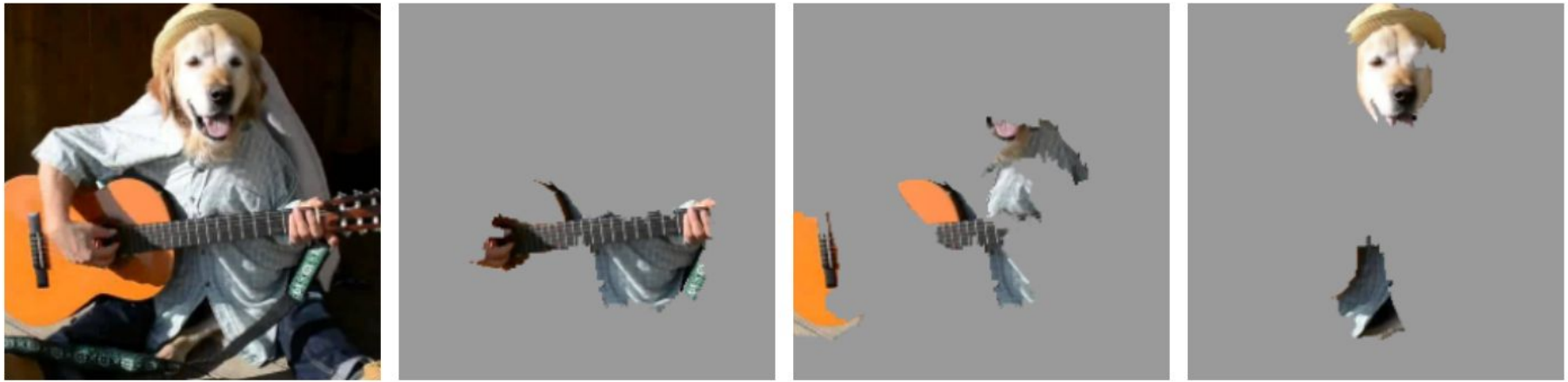


No connection to God or Atheism!

Example 2: GoogLeNet for Image Classification

- What can be an interpretable representation for image?
  - Super-pixels are used by the LIME paper

Example 2: GoogLeNet for Image Classification



(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google's Inception neural network. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$)**, "Acoustic guitar"** ($p = 0.24$) **and "Labrador"** ($p = 0.21$)

LIME: Pros

- Model-Agnostic
- Easy to use (libraries exist)
- Works with multiple data domains (text, images, etc.)
  - Interpretable data representations
- Selective and contrastive(?) explanations
- Formulated a fidelity metric

LIME: Cons

- Neighborhood - not easy to define
- More meaningful sampling
- Instability
  - Varied explanations for closeby samples (or, with different sampling)
- Explanations are manipulated - adversarial attacks on LIME & SHAP
- etc.

Reading exercise

- SHAP: Shapely values as importance [Lundberg & Lee 2017] [~25K citations]
- Marginal contribution of each feature towards the prediction
- Averaged over all the possible permutations

Next

- Counterfactuals
- Local Model-Agnostic Methods
  - Saliency Maps
  - Activation maximization
- Global Model-Agnostic Methods